

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Flight Status Agent | API-Workflow Domain (Travel Desk)

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — introduces live external flight status API integration

1. Problem Statement

Corporate travel desks and employees frequently exchange emails asking whether an upcoming flight is on time, delayed, or has changed gates — for example, an employee checking their flight status before heading to the airport for a business trip, or a travel coordinator confirming an executive's connecting flight. Today, staff manually check the airline's website or a flight-tracking app and reply by email — a small but repetitive task that delays responses, especially when travel volume peaks during conference season.

Design and build an Autonomous UiPath Agent, the AI Flight Status Agent, that reads an incoming email requesting a flight status check, extracts the flight number and date, calls a live Flight Status API workflow to fetch the current status, and independently replies with the result — without any human approval step.

2. System Prompt

You are an AI Flight Status Agent for the Corporate Travel Desk. Your goal is to analyze flight status request emails and provide an accurate, up-to-date flight status using a live flight tracking API.

Follow this reasoning process:

1. Read and understand the email content.
2. Classify it into one of the following:
 - Flight Status Request
 - Flight Delay/Cancellation Concern
 - Other
3. If it is Flight Status Request or Flight Delay/Cancellation Concern:
 - Extract the flight number and travel date
 - Call the Flight Status API workflow with the flight number and date
 - Decide if it is urgent (e.g., status shows 'Delayed' or 'Cancelled')
 - Generate a professional reply stating the current status, scheduled vs. estimated departure/arrival time, and gate information if available
4. If it is Other:
 - Skip processing

Rules:

- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details; if the flight number or date is missing, ask the requester to provide it instead of guessing
- Do not invent flight statuses – always use the API response
- Return output in structured format

3. User Prompt

Analyze the following flight status request email and provide the current flight status using the live flight tracking API:
{{flight_email_text}}

4. Input Schema

```
{  
  "flight_email_text": "string"  
}
```

5. Output Schema

```
{  
  "category": "string",  
  "flight_number": "string",  
  "travel_date": "string",  
  "current_status": "string",  
  "scheduled_time": "string",  
  "estimated_time": "string",  
  "gate": "string",  
  "urgency": "string",  
  "action": "string"  
}
```

6. Tool: API Workflow & Publish

Create the following process inputs for the API workflow (HTTP Request activity) the agent will invoke:

- flight_number (text) — e.g., "AI 202"
- travel_date (text) — format yyyy-MM-dd

Configure the workflow to call a live flight-status API endpoint (e.g., an HTTPS GET request to a flight-tracking provider) and parse the JSON response for status, scheduled/estimated times, and gate information.

7. Add Tool: Map Inputs

Agent Field	API Request Parameter
flight_number	flight_iata_code
travel_date	departure_date

8. Updated System Prompt — Tool Invocation Logic

```
If the email is classified as Flight Status Request or Flight
Delay/Cancellation Concern:
- Call the get_flight_status API workflow with flight_number and
  travel_date.
- Parse the API response for 'status', 'scheduled_time',
  'estimated_time', and 'gate'.
- If the returned status is 'Delayed' or 'Cancelled', set urgency =
  "High".
- Use the generate_reply GenAI activity to draft a professional reply
  email with the current status, scheduled vs. estimated time, and
  gate information – fully autonomously, with no manual approval
  required.

If the email is classified as Other:
- Take no further action.
```

9. Tools & Activities Summary

Tool / Activity	Type
get_flight_status	API Workflow (HTTP Request to a live flight status API)
generate_reply	GenAI Activity (LLM prompt-based text generation)
classify_request	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent classifies the request, calls the live flight status API, evaluates urgency, and composes the reply without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a system prompt that integrates a live external API into agent reasoning
- Configure an API Workflow (HTTP Request activity) with mapped input parameters
- Parse a JSON API response and derive urgency from the returned flight status
- Define structured input/output schemas for an API-integrated autonomous agent
- Publish and test a fully autonomous, API-driven agent in UiPath Agent Builder